

AI-assistance disclosure

AI-assistance disclosure — JHEP manuscript

Factual, and deliberately unflattering where that is accurate. No AI system is listed as an author, and none can be.

This extends `submission_candidate/ai_use_disclosure.md`, which covers the work predating this manuscript. Everything there still holds. What follows is what was added for the JHEP article.

What AI assistance did, for this manuscript

activity	extent
code generation	Substantial. The per-cell certificate driver, the read-only matrix aggregator, the execution-freeze and provenance tooling, the exact rational D10/Q10 closure, the source-corpus builder and the pytest counter were all AI-written.
test generation	All of them. 32 aggregator tests, 19 test-counter tests. Each was written by the same system that wrote the code they test.
debugging	Three defects were found and fixed. A flop-budget mismatch that presented as a scientific evaluation error; a cache key that collided two bibliographic records and served one paper's metadata under another's key; a test-counter fallback that would have miscounted a killed run as a clean one.
literature search	39 sources resolved against INSPIRE-HEP and Crossref, 98 references imported from the two core papers, 18 citing papers enumerated and assessed. This is a search with authoritative metadata, not a survey by someone who knows the field.
analysis	The exact rational D10 closure, the structural A10/G10/P10 argument, and the classification of which degree-10 statements are exposed to an exceptional prime.

activity	extent
manuscript drafting	Not yet begun at the time of writing. When it happens it will be AI-drafted and will require author rewriting.
claim narrowing	The prohibited-claims list, the novelty classifications, and the decision to exclude degree 12 were AI-proposed. They restrict what the paper says rather than extending it.

What AI assistance did not do

- **It did not verify anything independently.** Every check was written by the same system, in the same session, under the same assumptions. Agreement between an implementation and its own tests is not independent verification. The four adversarial referee simulations are subject to exactly this limitation and say so in their own text.
- **It did not establish novelty.** The prior-art finding that matters most in this manuscript — that Elamaram, Ferko and Scarlett already published the enumerate-evaluate-relate workflow — came from a citation sweep, and it narrowed the claim. But absence from a search is not evidence of absence, and recognising a prior formulation of the same statement requires knowing the field. Every novelty row remains provisional pending author judgement.
- **It did not decide authorship, and cannot.**
- **It did not obtain any approval.** No approval recorded anywhere in this repository was given by a human at the time of writing. The checklist in `audit/AUTHOR_APPROVAL_CHECKLIST.md` is empty by design.

One thing worth stating plainly

A referee is entitled to know how much of this was machine-written. The honest answer is: most of the code, all of the tests, all of the audit apparatus, and the analysis that produced the exact D10 result. What a human supplied is the research question, the mathematical design predating this work, the tensor-side implementation, and — when it happens — the judgement about whether any of it is worth publishing.

Human verification status

None has occurred. This is not a formality: three of the defects listed above were found only because the same system re-examined its own output under a different framing, and there is no reason to believe that process is exhaustive. Independent human review of the bridge derivation, the exact rational closure and the claim ledger is a prerequisite for submission, not a courtesy.

Final responsibility

Rests entirely with the human authors, who must be satisfied that every claim is one they would defend in front of a referee without reference to how it was produced.